

Choroidal vascular index in cystoid macular edema associated with retinitis pigmentosa

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ABSTRACT

Background: Retinitis pigmentosa (RP) is an inherited degenerative disease characterized by night blindness (nyctalopia), visual field defects, and varying degrees of vision loss. Choroid tissue has an essential role in the pathophysiology of many chorioretinal diseases. The choroidal vascularity index (CVI) is a choroidal parameter obtained as the ratio of the luminal choroidal area to the total choroidal area. The study aimed to compare the CVI of RP patients with and without CME with each other and with healthy individuals. **AU:** Please confirm that the provided email "ademoztel@hotmail.com" is the correct address for official communication, else provide an alternate e-mail address to replace the existing one..

Methods: A retrospective, comparative study of 76 eyes of 76 RP patients and 60 right eyes of 60 healthy subjects was conducted. The patients were divided into two groups: those with and those without cystoid macular edema (CME). The images were obtained using enhanced depth imaging optical coherence tomography (EDI-OCT). CVI was calculated by using the binarization method with ImageJ software.

Results: The mean CVI was significantly lower in RP patients compared to the control group (0.61 ± 0.05 and 0.65 ± 0.02 , respectively, $p < 0.01$). The mean CVI in RP patients with CME was significantly lower than in those without CME (0.60 ± 0.54 and 0.63 ± 0.35 , respectively, $p = 0.01$). The CVI was positively correlated with subfoveal choroidal thickness ($r = 0.74$, $p = 0.001$), central macular thickness ($r = 0.27$, $p < 0.001$) and visual acuity (logMAR) ($r = -0.23$, $p = 0.03$) in RP patients.

Conclusions: The CVI is lower in RP patients with CME than in patients without CME and in RP patients compared to healthy subjects, indicating an ocular vascular involvement in the pathophysiology of the disease and the pathogenesis of RP-associated cystoid macular edema.

1. Introduction

Retinitis pigmentosa (RP) is an inherited degenerative disease characterized primarily by the progressive loss of rod and concomitant cone cells. Around 40 years of age, night blindness (nyctalopia), visual field defects, and varying degrees of vision loss are typical in RP patients [1]. Considering the age of those affected by the disease, it is clear that it is an important public health issue that concerns society.

The choroid is one of the most intensely vascularized tissues in the human body. It plays an essential role in the oxygenation and nutrition of the outer retinal layers and retinal pigment epithelium [2]. Choroid tissue is key in the pathophysiology of many chorioretinal diseases. Studies have examined changes in the choroidal structure in RP and other hereditary retinal dystrophies [3–5]. Choroidal thickness is a

criterion used in evaluating the choroid in clinical studies. It reflects only the total choroidal vasculature, with no distinction between the stromal and luminal vascular components [6]. The choroidal vascular index (CVI) is a measure calculated by the ratio of the vascular luminal area of the choroid to the choroidal area, giving more detailed information about the choroidal vascular structure.

Although the pathogenesis of RP-associated cystoid macular edema (CME) cannot be clearly explained, disruption of the blood-retinal barrier, impaired pump function of the retinal pigment epithelium, Müller cell dysfunction, antiretinal antibodies, and vitreous traction have been suggested [7]. In our study, we aimed to compare the choroidal features of RP patients with and without CME with each other and with healthy individuals and to investigate the role of the choroid in the development of CME in RP patients.

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2. Materials and methods

Seventy-seven eyes of 77 patients diagnosed with retinitis pigmentosa who sought appointments at the retina clinic between January 2010 and January 2021 were selected consecutively. The patients were divided into two groups: those with CME (42 cases) and those without (35 cases). The control group included the right eye of 69 patients with best-corrected visual acuity of 1.0 and no systemic or ocular disease who came to the outpatient clinic for a routine eye examination. In cases with bilateral CME, the eye with excessive central macular thickness was included in the study, and the other eye was excluded. The right eye was selected for statistical analysis in RP patients without CME.

All patients underwent detailed ophthalmological examinations, including anterior segment examination with biomicroscopy, dilated fundus examination with 1% tropicamide, best corrected visual acuity with the Snellen chart, and intraocular pressure measurement with Goldmann applanation tonometry.

The diagnosis of RP was made with attenuation of retinal vessels, waxy optic disc pallor, and bone specula on dilated fundus examination and wide-angle fundus photography. In addition, family history, electroretinography, and visual field were used to support the diagnosis in cases deemed necessary. The genetic analysis was performed on a limited number of patients.

The patients underwent EDI-OCT scans using the EDI mode of SD-OCT (Spectralis, Heidelberg Engineering, Heidelberg, Germany). The macula was scanned using a seven horizontal line scan ($30^\circ \times 5^\circ$) centered on the fovea, with 100 frames averaged in each B-scan. Each scan was 8.9 mm in length and separated from the next by 240 μ m. The thickness profile module automatically displays CMT. The measurements were performed between 9:00 am and 04:00 pm. CME was defined as the presence of at least one intraretinal cyst in spectral domain optical coherence tomography (SD-OCT, Spectralis, Heidelberg Engineering).

Patients with concomitant systemic disease, ocular disease other than RP (including cataract), macular edema secondary epiretinal membrane, a history of trauma, eyes with hyperopia or myopia greater than 6 diopters according to auto-refractometer (Topcon KR8800 auto-refractometer/keratometer) values, and poor-quality OCT imaging, and those undergoing ocular surgery were excluded from the study.

Subfoveal choroidal thickness was measured manually using OCT software, with a perpendicular line the distance between the outer edge of the retinal pigment epithelium (RPE) and the choroid-sclera border [8]. ImageJ software (version 1.53; National Institutes of Health, Bethesda, MD, <http://imagej.nih.gov/ij/>) was used for CVI measurement. EDI-OCT images were first converted to 8-bit images. Niblack's automatic threshold tool delimited the luminal area (LA) and stromal area (SA) (Fig. 1). The total choroidal area (TCA) was determined by manually drawing the RPE's lower border and the choroid-sclera junction. The entire choroidal area on the EDI-OCT B-scan was measured. The areas of black pixels (LA) and white pixels (SA) were calculated. The choroidal vascularity index was calculated by dividing the luminal area (LA) by the total choroidal area (TCA). All measurements were made by two people and averaged.

The SPSS 20.0 (SPSS Inc., Chicago, IL, USA) program was used for data analysis. The conformity of the data to the normal distribution was evaluated with the Kolmogorov-Smirnov test. The paired t-test was used to compare normally distributed variables. One-way analysis of variance (ANOVA) was used for comparisons between groups, and the chi-square test was used for comparisons of qualitative data. The Pearson correlation test was used for normally distributed parameters, and the Spearman rank correlation test was used for non-normally distributed parameters. The possible association between BCVA and CVI with some parameters (central foveal thickness, subfoveal choroidal thickness, and age) was evaluated using linear regression analysis with a statistical threshold level of $p < 0.05$.

Medipol University Ethics Committee approval was obtained, and

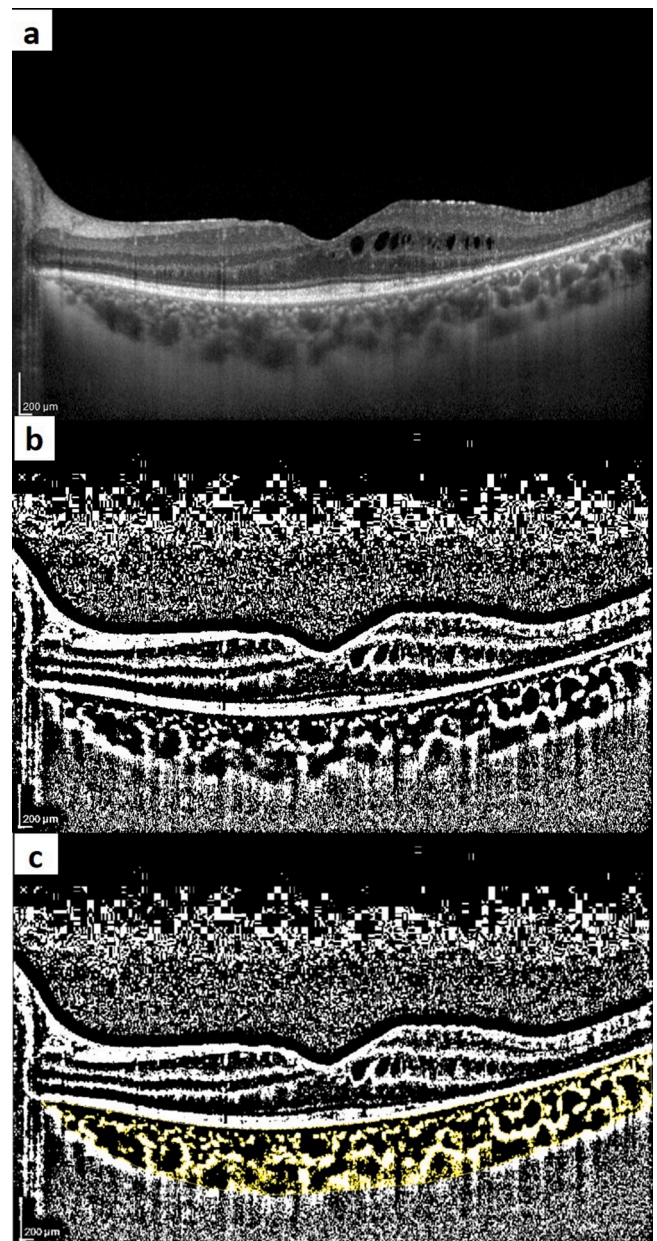


Fig. 1. CVI measurement technique in a healthy subject. a. Enhanced depth optical coherence tomography (EDI-OCT) image of the case b. Binary version of EDI-OCT images to determine the total choroidal area. c. The 'colour threshold' tool was used to select the black pixels representing the luminal area.

the Helsinki Declaration of Human Rights was adhered to throughout the study.

3. Results

The study was carried out on 146 eyes of 146 patients; 77 of the patients included in the study were RP patients, and 69 were in the control group. Of the subjects in the control group, 30 were female and 39 were male; their mean age was 45.33 ± 11.27 (15-63) years, and the mean spherical equivalent was -1.19 ± 2.83 D. The average age of RP patients was 39.24 ± 12.49 (14-58) years. Thirty of the RP patients were female and 47 were male, and CME was detected in 42 patients; in 26 it was bilateral. CME was not detected in 35 patients. Intraretinal cysts were primarily localized in the inner nuclear layer at a rate of 59.52%; they were localized in the outer nuclear layer at a rate of 7.14%.

Intraretinal cysts were localized in both inner and outer nuclear layers in 33.33% of the patients.

No statistically significant difference was observed between the RP and control groups' mean age and gender distributions. The mean best corrected visual acuity (BCVA) was 0.62 ± 0.29 (logMAR) in RP patients with CME and 0.53 ± 0.32 in patients without CME ($p=0.03$).

The mean subfoveal choroidal thickness was found to be thinner in RP patients ($247.27 \pm 74.69 \mu\text{m}$) than in the control group ($312.39 \pm 76.94 \mu\text{m}$) ($p<0.001$). The mean choroidal thickness was $251.08 \pm 59.34 \mu\text{m}$ in RP patients with CME and $244.91 \pm 83.39 \mu\text{m}$ in patients without CME ($p>0.05$).

A negative correlation was found between subfoveal choroidal thickness and age in RP patients and the control group (respectively: $r=-0.22$ $p=0.04$, $r=-0.27$ $p=0.04$), and higher age was associated with a thinner subfoveal choroidal thickness.

The mean CVI in eyes of RP patients was 0.61 ± 0.05 , which was significantly lower than in the control group (0.65 ± 0.02 , $p<0.01$). The mean CVI in RP patients with CME was significantly lower than that in RP patients without CME (0.60 ± 0.54 , 0.63 ± 0.35 , respectively, $p=0.01$).

The mean CFT was $262.98 \pm 92.62 \mu\text{m}$ in RP patients and $253.64 \pm 95.38 \mu\text{m}$ in the control group ($p>0.05$). The mean CFT was $326.80 \pm 113.64 \mu\text{m}$ in RP patients with CME and $186.40 \pm 53.81 \mu\text{m}$ in patients without CME. CFT had a significant correlation with BCVA in RP patients with CME ($r=0.44$ $p=0.02$) and without CME ($r=0.46$ $p=0.02$).

Using linear regression, we assessed the correlation between BCVA and variables such as CVI, CFT, and subfoveal choroidal thickness. We found that CVI ($r=-0.43$ $p=0.02$) and CFT ($r=0.24$ $p=0.02$) had significant correlations with BCVA; however, subfoveal choroidal thickness had no correlation with BCVA. We also assessed the correlation between the CVI and variables such as subfoveal choroidal thickness, CFT, and age. We found that subfoveal choroidal thickness ($r=0.74$, $p=0.001$) and CFT ($r=-0.27$, $p<0.001$) had significant correlations with CVI; however, age had no correlation with CVI.

4. Discussion

In this retrospective study, CVI and subfoveal choroidal thickness were measured to evaluate the vascular structure of the choroid in RP patients with and without CME and compare it with a control group of healthy subjects.

The possibility of a link between structural changes in the choroid and retinal outer segments and visual acuity has piqued the interest of researchers. Choroidal thickness, optical coherence tomography angiography (OCTA), and CVI have been evaluated in different studies. CVI is a recently defined choroidal parameter that reflects the vascular structure of the choroid and is obtained as the ratio of the luminal choroidal area to the total choroidal area. It is suggested that CVI is a more stable marker than subfoveal choroidal thickness when evaluating the structure of the choroid [9–12].

Ocular blood flow has been found to be reduced in RP patients [13]. This may be due to a decrease in choroidal blood flow secondary to a decrease in metabolic demand due to atrophy of the choroidal vasculature or outer and inner retinal layers; alternatively, retinal atrophy may have developed as a result of ischemia due to the decrease in the choroidal vascular structure. Toto et al. [14] also showed that RP patients' retinal and choriocapillaris vessel densities were significantly lower than those of healthy subjects. Atas et al [15], also reported that the CVI value was lower in RP patients than in healthy subjects. The histopathological features of RP patients are thickening of the vessel wall, sclerosis, and narrowing of the vessel lumen [16]. In our study, the CVI of RP patients was found to be lower than in the control group. Considering that CVI is obtained by dividing the luminal area by the total choroidal area (SA+LA), the increase in the stromal area and the decrease in the luminal area are better understood. In our study, the mean CVI in the healthy population was found to be 0.65 ± 0.02 , which

was similar to other studies. [12,17]

Reduced choroidal vascularity is a feature of retinal dystrophies, including RP, Stargardt disease, cone-rod dystrophy, Best disease, and Bietti crystalline dystrophy [5]. In a study conducted by Iovino et al. [18] on a total of 159 RP patients, it was reported that CVI was lower in RP patients with CME compared to those without CME [18]. Also, in our study, the mean CVI was lower in RP patients with CME compared to those without CME. These findings suggested that choroid vascular structure holds importance in the development of CME in eyes with RP. Inflammatory cells and cytokines were increased in the RP patients [19]. As we mentioned earlier regarding histopathological changes, the stroma of the choroidal vessels thickens and the vessel lumen narrows, possibly with the contribution of the inflammatory process, resulting in a decrease in CVI. In this context, it can be speculated that CVI is an indirect indicator of inflammation and disease progression. The low CVI in RP patients with CME may indicate the development of CME due to the advanced stage of the disease and inflammation.

The mean subfoveal CT is reduced in RP patients compared to healthy controls [8]. Iovino et al. [18] report that the mean subfoveal CT was increased in RP patients with CME compared with RP patients without CME [18]. According to their hypotheses, the thicker choroid in RP patients with CME may be caused by increased choroidal blood flow secondary to intraocular inflammation, and as a result of this mechanism, CME occurs. In our study, there is no statistical difference in the mean choroidal thickness of RP patients with and without CME. We found that as CVI increased, BCVA was better; however, we did not find a significant relationship between subfoveal choroidal thickness and BCVA. While some other studies have reported a relationship between choroidal thickness and BCVA in RP patients [18,20], other studies have not. [8,21] While a significant correlation was found between BCVA and CVI, it was not found between subfoveal choroidal thickness; CVI may be a better marker of choroidal function than subfoveal choroidal thickness in RP patients. The increase in the choroidal stromal area as a result of the thickening of the vessel wall and the decrease in the luminal area as a result of the narrowing of the vessel lumen may attenuate each other when the total choroidal area is calculated. This may be the reason that CVI is a better marker than subfoveal choroidal thickness for showing disease progression.

Our study found that BCVA decreased as CFT increased in RP patients with CME, and BCVA decreased as CFT decreased in patients without CFT. A study in patients with RP with CME reported an association between retinal thickening from CME and poor visual acuity [22]. Another study in patients with RP but without CME reported a central retinal thinning secondary to retinal cell loss associated with poor visual acuity [23].

Agrawal et al. [12] reported a correlation between age and CVI. In other studies conducted in healthy populations and RP patients, a correlation was found between CT and age but not with CVI [18,24]. Similarly, no significant correlation was found between age and CVI in our study. Based on these results, CVI may be less affected by age than by CT.

CME was detected in 54.6% of the RP patients included in our study and bilaterally in 62.3%. There are studies reporting different percentages of CME and bilaterality [25–27]. The reported prevalence values differ according to the definition of CME and the characteristics of the RP patient group included in the study.

The retrospective nature of the study and the lack of genetic analysis in addition to clinical diagnosis are among the weaknesses of the study. Although the measurements were taken during working hours, not all shots were taken at the same time of day. Considering that CVI shows diurnal variation, it may have affected the results [24]. CVI could be affected by alterations in regional signal density due to hypo- or hyper-transmission defects because of RPE changes [28]. The inability to eliminate the potential backscattering and shadowing caused by RPE on CVI is another limitation of the study.

In conclusion, choroidal vascular structural changes may have

played an important role in the development of CME in RP eyes, and CVI may be a better marker than choroidal thickness for showing choroidal vascular structure and disease progression in RP patients. The evaluation of choroidal features may provide significant information that could be useful to understanding the pathophysiology of the disease and potential treatment strategies for RP patients.

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CRediT authorship contribution statement

Adem Tellioglu: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rukiye Aydin:** .

Declaration of Competing Interest

The authors declare that they have no conflict of interest.

References

- [1] D.T. Hartong, E.L. Berson, T.P. Dryja, Retinitis pigmentosa, *Lancet* (London, England) 368 (2006) 1795–1809.
- [2] R.A. Linsenmeier, L. Padnick-Silver, Metabolic dependence of photoreceptors on the choroid in the normal and detached retina, *Invest. Ophthalmol. Vis. Sci.* 41 (2000) 3117–3123.
- [3] F. Giansanti, G. Vicini, A. Sodi, et al., Optical coherence tomography angiography for the evaluation of retinal and choroidal vasculature in retinitis pigmentosa: a monocentric experience, *Diagnostics* (Basel, Switzerland) (2022) 12.
- [4] C.K. Yoon, K. Bae, H.G. Yu, Longitudinal microstructure changes of the retina and choroid in retinitis pigmentosa, *Am. J. Ophthalmol.* 241 (2022) 149–159.
- [5] X. Wei, C. Mishra, N.B. Kannan, et al., Choroidal structural analysis and vascularity index in retinal dystrophies, *Acta Ophthalmol.* 97 (2019) e116–e121.
- [6] F. Pichi, K. Aggarwal, P. Neri, et al., Choroidal biomarkers, *Indian J. Ophthalmol.* 66 (2018) 1716–1726.
- [7] S. Strong, G. Liew, M. Michaelides, Retinitis pigmentosa-associated cystoid macular oedema: pathogenesis and avenues of intervention, *Br. J. Ophthalmol.* 101 (2017) 31–37.
- [8] A. Sodi, C. Lenzetti, V. Murro, et al., EDI-OCT evaluation of choroidal thickness in retinitis pigmentosa, *Eur. J. Ophthalmol.* 28 (2018) 52–57.
- [9] M. Egawa, Y. Mitamura, M. Niki, et al., Correlations between choroidal structures and visual functions in eyes with Retinitis pigmentosa, *Retina* 39 (2019) 2399–2409.
- [10] R. Tan, R. Agrawal, S. Taduru, A. Gupta, K. Vupparaboina, J. Chhablani, Choroidal vascularity index in retinitis pigmentosa: an OCT study, *Ophthalmol. Surg. Lasers Imaging Retina* 49 (2018) 191–197.
- [11] R. Mastropasqua, R. D'Aloisio, C. De Nicola, et al., Widefield swept source OCTA in Retinitis Pigmentosa, *Diagnostics* (Basel, Switzerland) (2020) 10.
- [12] R. Agrawal, P. Gupta, K.-A. Tan, C.M.G. Cheung, T.-Y. Wong, C.-Y. Cheng, Choroidal vascularity index as a measure of vascular status of the choroid: measurements in healthy eyes from a population-based study, *Sci. Rep.* 6 (2016) 21090.
- [13] J.E. Grunwald, A.M. Maguire, J. Dupont, Retinal hemodynamics in retinitis pigmentosa, *Am. J. Ophthalmol.* 122 (1996) 502–508.
- [14] L. Toto, E. Borrelli, R. Mastropasqua, et al., Macular features in retinitis pigmentosa: correlations among ganglion cell complex thickness, capillary density, and macular function, *Invest. Ophthalmol. Vis. Sci.* 57 (2016) 6360–6366.
- [15] F. Ataş, M. Kayabaşı, A.O. Saatci, Vessel density and choroidal vascularity index in patients with Bietti crystalline dystrophy and retinitis pigmentosa, *Photodiagnosis Photodyn. Ther.* 40 (2022), 103181.
- [16] A.H. Milam, Z.Y. Li, R.N. Fariss, Histopathology of the human retina in retinitis pigmentosa, *Prog. Retin. Eye Res.* 17 (1998) 175–205.
- [17] M.Z. Karslioglu, C. Kesim, O. Yucel, et al., Choroidal vascularity index in pseudoexfoliative glaucoma, *Int. Ophthalmol.* 41 (2021) 4197–4208.
- [18] C. Iovino, A. Au, A. Hilely, et al., Evaluation of the Choroid in Eyes With Retinitis Pigmentosa and Cystoid Macular Edema, *Invest. Ophthalmol. Vis. Sci.* 60 (2019) 5000–5006.
- [19] N. Yoshida, Y. Ikeda, S. Notomi, et al., Clinical evidence of sustained chronic inflammatory reaction in retinitis pigmentosa, *Ophthalmology* 120 (2013) 100–105.
- [20] L.N. Ayton, R.H. Guymer, C.D. Luu, Choroidal thickness profiles in retinitis pigmentosa, *Clin. Experiment. Ophthalmol.* 41 (2013) 396–403.
- [21] D.S. Dhoot, S. Huo, A. Yuan, et al., Evaluation of choroidal thickness in retinitis pigmentosa using enhanced depth imaging optical coherence tomography, *Br. J. Ophthalmol.* 97 (2013) 66–69.
- [22] G. Liew, A.T. Moore, P.D. Bradley, A.R. Webster, M. Michaelides, Factors associated with visual acuity in patients with cystoid macular oedema and Retinitis Pigmentosa, *Ophthalmic Epidemiol.* 25 (2018) 183–186.
- [23] M.A. Sandberg, R.J. Brockhurst, A.R. Gaudio, E.L. Berson, The association between visual acuity and central retinal thickness in retinitis pigmentosa, *Invest. Ophthalmol. Vis. Sci.* 46 (2005) 3349–3354.
- [24] S.R. Singh, M.A. Rasheed, A. Goud, N.K. Sahoo, K.K. Vupparaboina, J. Chhablani, Diurnal variation in subfoveal and peripapillary choroidal vascularity index in healthy eyes, *Indian J. Ophthalmol.* 67 (2019) 1667–1672.
- [25] G. Liew, S. Strong, P. Bradley, et al., Prevalence of cystoid macular oedema, epiretinal membrane and cataract in retinitis pigmentosa, *Br. J. Ophthalmol.* 103 (2019) 1163–1166.
- [26] C.A. Adackapara, J.S. Sunness, C.W. Dibernardo, B.M. Melia, G. Dagnelie, Prevalence of cystoid macular edema and stability in oct retinal thickness in eyes with retinitis pigmentosa during a 48-week lutein trial, *Retina* 28 (2008) 103–110.
- [27] F. Testa, S. Rossi, R. Colucci, et al., Macular abnormalities in Italian patients with retinitis pigmentosa, *Br. J. Ophthalmol.* 98 (2014) 946–950.
- [28] H. Zhou, J. Lu, K. Chen, et al., Mitigating the effects of choroidal hyper- and hypo-transmission defects on choroidal vascularity index assessments using optical coherence tomography, *Quant. Imaging Med. Surg.* 12 (2022) 2932–2946.